

FAHIM DALVI

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EDUCATION

Stanford University

Masters of Science in Computer Science

Artificial Intelligence and Machine Learning

📅 2016

🎓 GPA: 4.0

Carnegie Mellon University

Bachelors of Science in Computer Science

Minor in Mathematics

📅 2014

🎓 GPA: 3.97 (University Honors)

EXPERIENCE

Senior Software Engineer 📅 July 2016 – Ongoing

Qatar Computing Research Institute

- Pre-trained **Fanar** and **Fanar 2.0**, the **best in-class Arabic Large Language Models** on the Open Arabic LLM leaderboard
- Developed award-winning Arabic **Machine Translation** models
- Led the software engineering efforts to develop **toolkits, libraries and scalable APIs** for the translation & interpretation team
- Developed research assets to simplify **interpretation** of deep NLP models
- Co-authored over 40 research papers in areas of machine translation and **explainable AI** in top tier conferences and journals
- Co-taught **Deep Learning** courses to 100+ students

Co-Founder

📅 March 2015 – March 2016

Problemia

Technical lead for designing and developing an **educational platform for teachers**. Managed a team of four and defined the technical direction of the platform

3D Content Creator

📅 Summer 2012

Williams F1

Created **interactive 3D content** for a simulator tuned for Qatar's driving environment

Research Intern

📅 Summer 2012

Robotics Institute, Carnegie Mellon

Designed and implemented a **user interface to analyze results** for a highly distributed data collection system, the AirBoats project

PROJECTS

Fanar

📍 QCRI 📅 2024 – Present

Pretraining Lead for **Fanar** text models, Qatar's Arabic-centric LLM. Contributed to **modeling, benchmarking** and **deployment** efforts of this National endeavor.

🔗 fanar.qcri.org

LLMeBench

📍 QCRI 📅 2023 – Present

- Created a framework for efficient benchmarking of **Large Language models** across a large suite of tasks
- Cloned over **2000 times** over the course of six months by the community, with over **300k samples** evaluated within QCRI

🔗 llmebench.qcri.org

NeuroX: Explainable AI

📍 QCRI 📅 2018 – Present

- Analyzed and interpreted Machine Translation and **Large Language models** like BERT, RoBERTa, GPT-2 and ALBERT
- Developed a **neural network diagnostic toolkit** for finding and analyzing important neurons for natural language processing models

🔗 neurox.qcri.org

Misinformation Detector

📍 QCRI 📅 2020

Trained and deployed models to detect misinformation in tweets related to the **COVID-19** pandemic

Text-to-Speech

📍 QCRI 📅 2019 – Present

- Developed a backend and associated REST API to host and run text-to-speech models with **accelerated inference** capabilities
- Created an **ML-Ops pipeline** for seamless deployment of models

🔗 tts.qcri.org

Live Speech Translation

📍 QCRI 📅 2017 - 2020

Designed and developed a live Arabic ↔ English transcription and translation app with a robust backend enabling live broadcastable sessions. Served **100+ hours** of live speech translation sessions

🔗 st.qcri.org

Machine Translation API

📍 QCRI 📅 2016 - Present

Developed a **distributed backend** to manage multiple machine translation engines built at QCRI and a simple to use user-facing REST API, serving over 165 million requests from 40+ countries. Packaged translation engines for **on-premise deployments**.

🔗 mt.qcri.org

SKILLS

PyTorch

Keras

Tensorflow

NodeJS

Flask

React

Docker

Python

C++

Javascript

Deep Learning

Machine Learning

Unix

Web Applications

Backend Development

Scripting

NLP Processing

Containerization

Code & Model Versioning

Testing

TEACHING

Deep Learning for NLP

Carnegie Mellon University

Guest Lecturer

Qatar

October 2022

Deep Learning for NLP

Carnegie Mellon University

Guest Lecturer

Qatar

October 2020

Deep Learning for NLP

University of Duisberg-Essen

Co-Lecturer

Germany

September 2019

Deep Learning for NLP

International Spring School in Advanced Language Engineering (ISSALE)

Co-Lecturer

Sri Lanka

March 2019

Deep Learning for Machine Translation

Computational Linguistics Fall School

Co-Lecturer

Germany

September 2017

Organized by the German Linguistic Society (Deutsche Gesellschaft für Sprachwissenschaft, DGfS).

ACHIEVEMENTS

Best Audience Experience

BBC NewsHack 2017

Best Arabic machine translation system

International Conference on Spoken Language Translation (IWSLT) 2016

Best MYO hack

Hack Overflow, Stanford 2015

Hamad Bin Khalifa University President's Award

Qatar Foundation 2014

Outstanding Academic Achievement

Carnegie Mellon 2014

Senior Student Leadership Award

Carnegie Mellon 2014

1st Place

Oman Collegiate Programming Competition 2012

PUBLICATIONS

- Team, F. (2026). Fanar 2.0: Arabic generative ai stack. <https://arxiv.org/abs/2603.16397>
- Boughorbel, S., Dalvi, F., Durrani, N., & Hawasly, M. (2025). Beyond the leaderboard: Understanding performance disparities in large language models via model diffing. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 31360–31371. <https://doi.org/10.18653/v1/2025.emnlp-main.1598>
- Durrani, N., Mousi, B., & Dalvi, F. (2025). Editing across languages: A survey of multilingual knowledge editing. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 15906–15918. <https://doi.org/10.18653/v1/2025.emnlp-main.803>
- Mousi, B., Durrani, N., & Dalvi, F. (2025). An exploration of knowledge editing for Arabic. *Proceedings of The Third Arabic Natural Language Processing Conference*, 417–424. <https://doi.org/10.18653/v1/2025.arabicnlp-main.34>
- Ersoy, A., Mousi, B. A., Chowdhury, S. A., Alam, F., Dalvi, F. I., & Durrani, N. (2025). From Words to Waves: Analyzing Concept Formation in Speech and Text-Based Foundation Models. *Interspeech 2025*, 241–245. <https://doi.org/10.21437/Interspeech.2025-2180>
- Team, F. (2025). Fanar: An arabic-centric multimodal generative ai platform. <https://arxiv.org/abs/2501.13944>
- Mousi, B., Durrani, N., Ahmad, F., Hasan, M. A., Hasanain, M., Kabbani, T., Dalvi, F., Chowdhury, S. A., & Alam, F. (2025, January). AraDiCE: Benchmarks for dialectal and cultural capabilities in LLMs. In O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. D. Eugenio, & S. Schockaert (Eds.), *Proceedings of the 31st international conference on computational linguistics* (pp. 4186–4218). Association for Computational Linguistics. <https://aclanthology.org/2025.coling-main.283/>
- Yu, X., Dalvi, F., Durrani, N., Nouri, M., & Sajjad, H. (2024, November). Latent concept-based explanation of NLP models. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 12435–12459). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.emnlp-main.692>
- Mousi, B., Durrani, N., Dalvi, F., Hawasly, M., & Abdelali, A. (2024, August). Exploring alignment in shared cross-lingual spaces. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 6326–6348). Association for Computational Linguistics. <https://aclanthology.org/2024.acl-long.344>
- Dalvi, F., Hasanain, M., Boughorbel, S., Mousi, B., Abdaljalil, S., Nazar, N., Abdelali, A., Chowdhury, S. A., Mubarak, H., Ali, A., Hawasly, M., Durrani, N., & Alam, F. (2024). LLMeBench: A flexible framework

for accelerating llms benchmarking.

- Hawasly, M., **Dalvi, F.**, & Durrani, N. (2024). Scaled-up discovery of latent concepts in deep NLP models. *18th Conference of the European Chapter of the Association for Computational Linguistics*. <https://openreview.net/forum?id=mdMafLCRNq>
- Abdelali, A., Mubarak, H., Chowdhury, S. A., Hasanain, M., Mousi, B., Boughorbel, S., Abdaljalil, S., Kheir, Y. E., Izham, D., **Dalvi, F.**, Hawasly, M., Nazar, N., Elshahawy, Y. I., Ali, A., Durrani, N., Milic-Frayling, N., & Alam, F. (2024). LArabench: Benchmarking arabic AI with large language models. *18th Conference of the European Chapter of the Association for Computational Linguistics*. <https://openreview.net/forum?id=ssQdvBLc5x>
- Fan, Y., **Dalvi, F.**, Durrani, N., & Sajjad, H. (2023). Evaluating neuron interpretation methods of NLP models. *Thirty-seventh Conference on Neural Information Processing Systems*. <https://openreview.net/forum?id=YiwMpyMdPX>
- Mousi, B., Durrani, N., & **Dalvi, F.** (2023, December). Can LLMs facilitate interpretation of pre-trained language models? In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 3248–3268). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.emnlp-main.196>
- **Dalvi, F.**, Sajjad, H., & Durrani, N. (2023). NeuroX library for neuron analysis of deep NLP models. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 226–234. <https://doi.org/10.18653/v1/2023.acl-demo.21>
- **Dalvi, F.**, Durrani, N., Sajjad, H., Jaban, T., Husaini, M., & Abbas, U. (2023). NxPlain: A web-based tool for discovery of latent concepts. *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, 75–83. <https://doi.org/10.18653/v1/2023.eacl-demo.10>
- Alam, F., **Dalvi, F.**, Durrani, N., Sajjad, H., Khan, A. R., & Xu, J. (2023). ConceptX: A framework for latent concept analysis. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(13), 16395–16397. <https://doi.org/10.1609/aaai.v37i13.27057>
- Sajjad, H., **Dalvi, F.**, Durrani, N., & Nakov, P. (2023). On the effect of dropping layers of pre-trained transformer models. *Computer Speech & Language*, 77, 101429. <https://doi.org/https://doi.org/10.1016/j.csl.2022.101429>
- Durrani, N., Sajjad, H., **Dalvi, F.**, & Alam, F. (2022). On the transformation of latent space in fine-tuned NLP models. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 1495–1516. <https://doi.org/10.18653/v1/2022.emnlp-main.97>
- Abdelali, A., Durrani, N., **Dalvi, F.**, & Sajjad, H. (2022). Post-hoc analysis of Arabic transformer models. *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, 91–103. <https://doi.org/10.18653/v1/2022.blackboxnlp-1.8>
- Abdelali, A., Durrani, N., Demiroglu, C., **Dalvi, F.**, Mubarak, H., & Darwish, K. (2022). NatiQ: An end-to-end text-to-speech system for Arabic. *Proceedings of the The Seventh Arabic Natural Language Processing Workshop (WANLP)*, 394–398. <https://doi.org/10.18653/v1/2022.wanlp-1.38>
- Sajjad, H., Durrani, N., & **Dalvi, F.** (2022). Neuron-level interpretation of deep NLP models: A survey. *Transactions of the Association for Computational Linguistics*, 10, 1285–1303. https://doi.org/10.1162/tacl_a_00519
- Sajjad, H., Alam, F., **Dalvi, F.**, & Durrani, N. (2022). Effect of post-processing on contextualized word representations. *Proceedings of the 29th International Conference on Computational Linguistics*.
- Sajjad, H., Durrani, N., **Dalvi, F.**, Alam, F., Khan, A., & Xu, J. (2022). Analyzing encoded concepts in transformer language models. *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 3082–3101. <https://doi.org/10.18653/v1/2022.naacl-main.225>
- ***Dalvi, F.**, *Khan, A. R., Alam, F., Durrani, N., Xu, J., & Sajjad, H. (2022). Discovering latent concepts learned in BERT. *International Conference on Learning Representations*. <https://openreview.net/forum?id=POTMtpYI1xH>
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- Alam, F., Shaar, S., **Dalvi, F.**, Sajjad, H., Nikolov, A., Mubarak, H., Da San Martino, G., Abdelali, A., Durrani, N., Darwish, K., Al-Homaid, A., Zaghouani, W., Caselli, T., Danoe, G., Stolk, F., Bruntink, B., & Nakov, P. (2021). Fighting the COVID-19 infodemic: Modeling the perspective of journalists, fact-checkers, social media platforms, policy makers, and the society. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 611–649. <https://aclanthology.org/2021.findings-emnlp.56>
- Alam, F., **Dalvi, F.**, Shaar, S., Durrani, N., Mubarak, H., Nikolov, A., Da San Martino, G., Abdelali, A., Sajjad, H., Darwish, K., & Nakov, P. (2021). Fighting the covid-19 infodemic in social media: A holistic perspective and a call to arms. *Proceedings of the International AAAI Conference on Web and Social Media*, 15(1), 913–922. <https://ojs.aaai.org/index.php/ICWSM/article/view/18114>
- Sajjad, H., Kokhlikyan, N., **Dalvi, F.**, & Durrani, N. (2021). Fine-grained interpretation and causation analysis in deep NLP models. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Tutorials*, 5–10. <https://www.aclweb.org/anthology/2021.naacl-tutorials.2>
- Sajjad, H., Abdelali, A., Durrani, N., & **Dalvi, F.** (2020). AraBench: Benchmarking dialectal Arabic-English

- machine translation. *Proceedings of the 28th International Conference on Computational Linguistics*, 5094–5107. <https://doi.org/10.18653/v1/2020.coling-main.447>
- Durrani, N., Sajjad, H., **Dalvi, F.**, & Belinkov, Y. (2020). Analyzing individual neurons in pre-trained language models. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 4865–4880. <https://doi.org/10.18653/v1/2020.emnlp-main.395>
 - **Dalvi, F.**, Sajjad, H., Durrani, N., & Belinkov, Y. (2020). Analyzing redundancy in pretrained transformer models. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 4908–4926. <https://doi.org/10.18653/v1/2020.emnlp-main.398>
 - Wu, J., Belinkov, Y., Sajjad, H., Durrani, N., **Dalvi, F.**, & Glass, J. (2020). Similarity analysis of contextual word representation models. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4638–4655. <https://doi.org/10.18653/v1/2020.acl-main.422>
 - *Belinkov, Y., *Durrani, N., **Dalvi, F.**, Sajjad, H., & Glass, J. (2020). On the linguistic representational power of neural machine translation models. *Computational Linguistics*, 46(1), 1–52. https://doi.org/10.1162/coli_a_00367
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 - Durrani, N., **Dalvi, F.**, Sajjad, H., Belinkov, Y., & Nakov, P. (2019). One size does not fit all: Comparing NMT representations of different granularities. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 1504–1516. <https://doi.org/10.18653/v1/N19-1154>
 - Bau, A., Belinkov, Y., Sajjad, H., Durrani, N., **Dalvi, F.**, & Glass, J. (2019). Identifying and controlling important neurons in neural machine translation. *International Conference on Learning Representations*. <https://openreview.net/forum?id=H1z-PsR5KX>
 - ***Dalvi, F.**, *Durrani, N., *Sajjad, H., Belinkov, Y., Bau, A., & Glass, J. (2019). What is one grain of sand in the desert? analyzing individual neurons in deep nlp models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01), 6309–6317. <https://doi.org/10.1609/aaai.v33i01.33016309>
 - **Dalvi, F.**, Nortonsmith, A., Bau, A., Belinkov, Y., Sajjad, H., Durrani, N., & Glass, J. (2019). NeuroX: A toolkit for analyzing individual neurons in neural networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01), 9851–9852. <https://doi.org/10.1609/aaai.v33i01.33019851>
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 - **Dalvi, F.**, *Durrani, N., Sajjad, H., & Vogel, S. (2018). Incremental decoding and training methods for simultaneous translation in neural machine translation. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, 493–499. <https://doi.org/10.18653/v1/N18-2079>
 - Barrón-Cedeño, A., Da San Martino, G., Zhang, Y., Ali, A. M., & **Dalvi, F.** (2018). Qlusty: Quick and dirty generation of event videos from written media coverage. *NewsIR@ECIR, 2019*, 27–32.
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 - Durrani, N., & **Dalvi, F.** (2017). Continuous space re-ordering models for phrase-based mt. *International Workshop on Spoken Language Translation*.
 - **Dalvi, F.**, Durrani, N., Sajjad, H., Belinkov, Y., & Vogel, S. (2017). Understanding and improving morphological learning in the neural machine translation decoder. *Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 142–151. <https://www.aclweb.org/anthology/I17-1015>
 - Belinkov, Y., Màrquez, L., Sajjad, H., Durrani, N., **Dalvi, F.**, & Glass, J. (2017). Evaluating layers of representation in neural machine translation on part-of-speech and semantic tagging tasks. *Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 1–10. <https://www.aclweb.org/anthology/I17-1001>
 - Sajjad, H., **Dalvi, F.**, Durrani, N., Abdelali, A., Belinkov, Y., & Vogel, S. (2017). Challenging language-dependent segmentation for Arabic: An application to machine translation and part-of-speech tagging. *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 601–607. <https://doi.org/10.18653/v1/P17-2095>
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